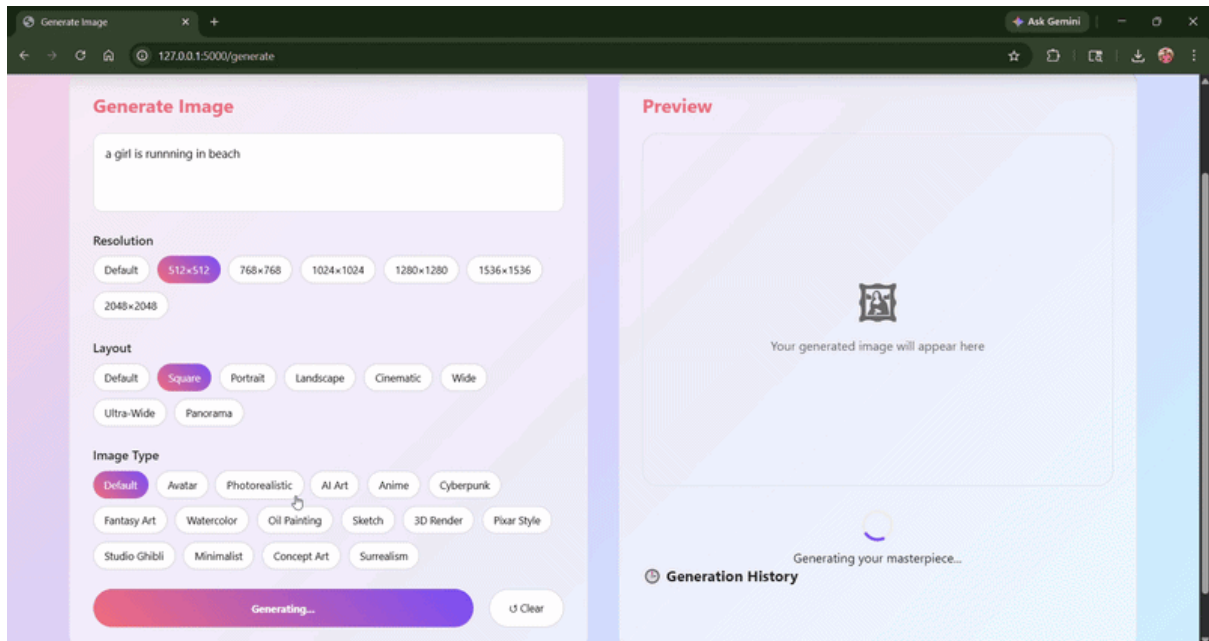


🌟 SDXL Studio — AI Image Generator

Turn text into stunning, high-resolution AI-generated images using **Stable Diffusion XL (SDXL)**.

🚀 Built with Flask, PyTorch, Diffusers, and Ngrok.

🖼️ Demo (GIF)



🌟 Features

- 🗨️ **Stable Diffusion XL (SDXL)** for high-quality image generation
- ⚡ Fast GPU-based inference
- 🎯 Prompt-based image creation
- 💡 AI-powered prompt suggestions
- 🖼️ Multiple styles (Anime, Photorealistic, Fantasy, etc.)
- 📐 Adjustable resolution & layout
- 📋 Generation history preview

🏗️ Architecture

1. Frontend (Flask UI)

- User prompt input

- Style & resolution selection
- Image preview

2. Backend (AI Server)

- SDXL model for image generation
- GPT model for prompt suggestions

3. Workflow

Prompt → CLIP Encoder → Latent Noise → U-Net Denoising → VAE Decode → Image

Tech Stack

- Python 
 - Flask 
 - Stable Diffusion XL 
 - HuggingFace Diffusers
 - PyTorch 
 - Transformers (GPT2)
 - Ngrok (API tunneling)
-

Installation

1 Clone Repo

```
git clone https://github.com/your-username/sd-xl-studio.git
```

```
cd sd-xl-studio
```

2 Install Dependencies

```
pip install -r requirements.txt
```

3 Run Backend (Colab / GPU)

- Start SDXL API server
- Copy Ngrok URL

4 Update API URLs

```
GENERATE_API = "YOUR_NGROK_URL/api/generate"
```

```
RECOMMEND_API = "YOUR_NGROK_URL/api/recommend"
```

5 Run Frontend

python app.py

API Endpoints

◆ Generate Image

POST /api/generate

Request:

```
{  
  "prompt": "A futuristic city at sunset"  
}
```

Response:

```
{  
  "image": "base64_string"  
}
```

◆ Prompt Recommendations

POST /api/recommend

Project Structure

sdxl-studio/

```
|  
|  
| └─ app.py          # Flask frontend  
| └─ templates/  
|   | └─ index.html  
|   | └─ generate.html  
|   | └─ about.html  
|   | └─ architecture.html  
|  
| └─ static/  
|   | └─ css/  
|   | └─ js/
```

```
|  ├── images/
|
|  ├── assets/
|  │   ├── demo.gif    # Demo GIF
|
|  └── README.md
```